

PRACTICA N°4

PRUEBAS DE RENDIMIENTO CON APACHE JMeter

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Materia: Verificación y Validación de Software – INF 780

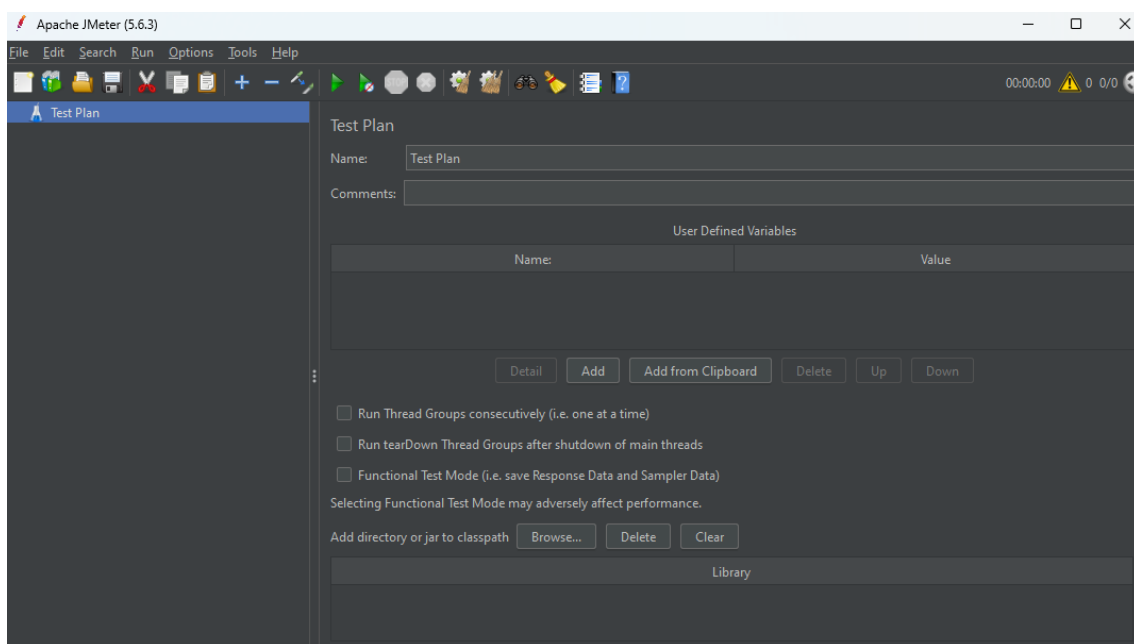
Fecha: 01/06/26

Parte 1 — Instalación y verificación

1. Descargue Apache JMeter 5.6.x desde jmeter.apache.org (requiere Java 8 o superior).
2. Descomprima y ejecute bin/jmeter (Windows: jmeter.bat; Linux/Mac: ./jmeter).
3. Confirme que la API está levantada y respondiendo en localhost:3000.
4. Capture la pantalla de inicio de JMeter y la versión de Java (java -version).

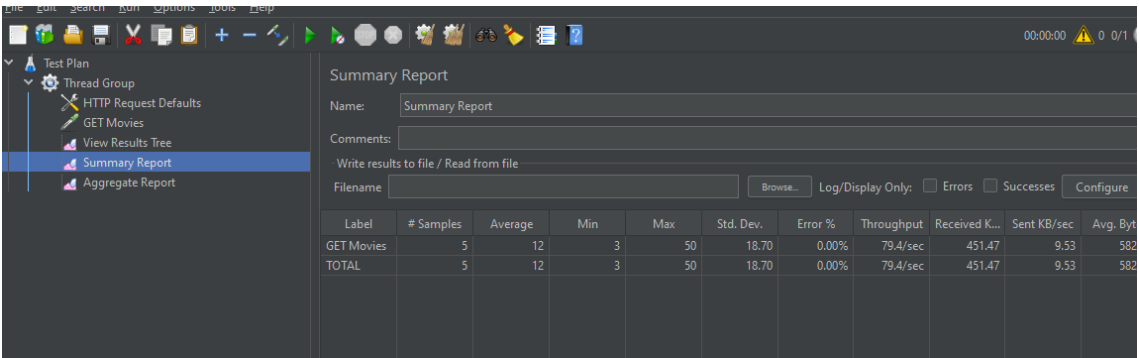
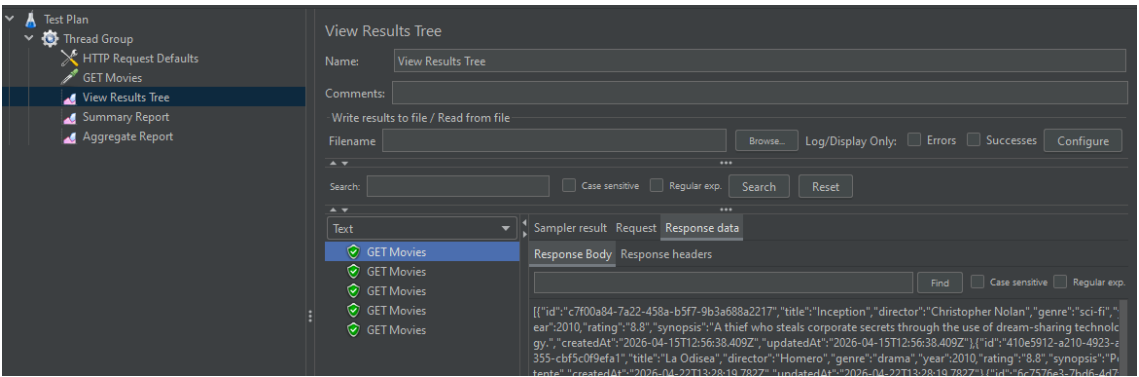
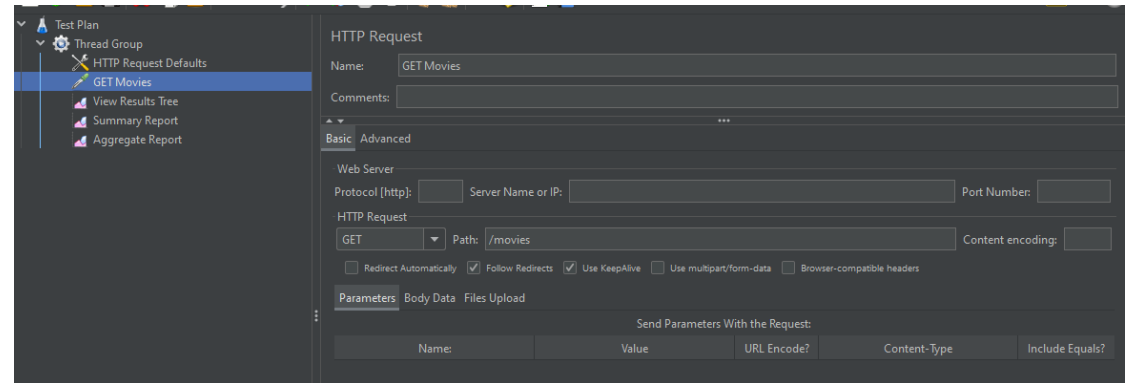
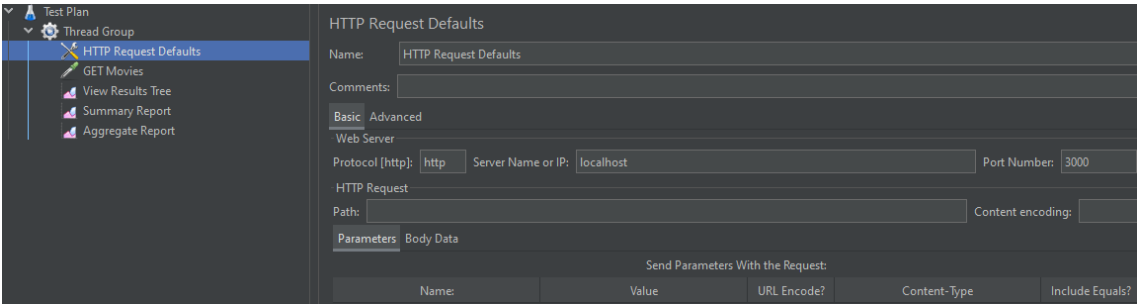
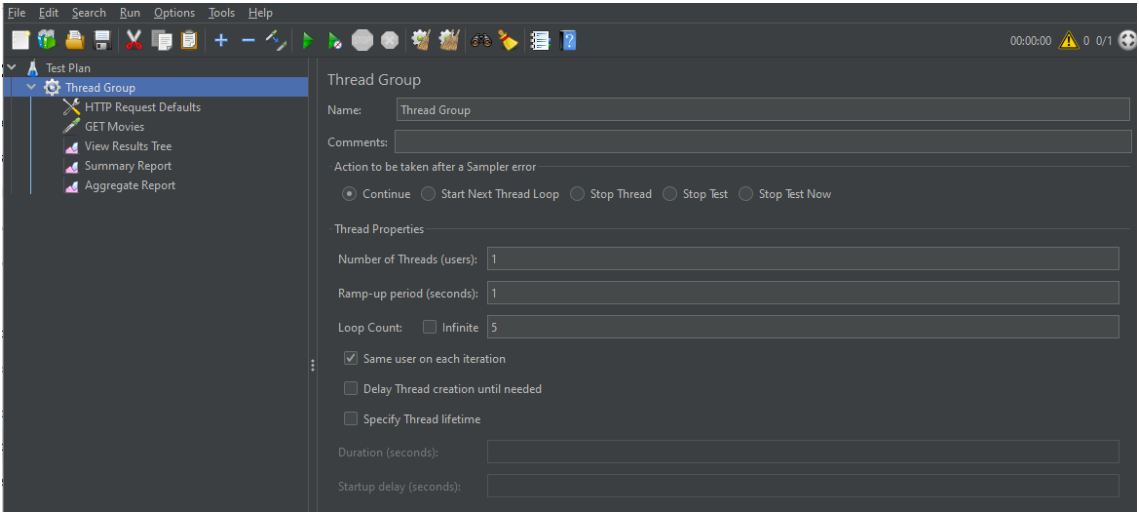
```
C:\Users\pc>java -version
openjdk version "21.0.11" 2026-04-21 LTS
OpenJDK Runtime Environment Temurin-21.0.11+10 (build 21.0.11+10-LTS)
OpenJDK 64-Bit Server VM Temurin-21.0.11+10 (build 21.0.11+10-LTS, mixed mode, sharing)

C:\Users\pc>
```



Parte 2 — Plan base (smoke test)

1. Cree un Test Plan y agregue un HTTP Request Defaults (Server Name: localhost, Port: 3000).
2. Agregue un Thread Group: 1 hilo, ramp-up 1 s, loops 5.
3. Agregue un HTTP Request GET /movies.
4. Agregue los listeners: View Results Tree, Summary Report y Aggregate Report.
5. Ejecute y verifique que todas las peticiones son exitosas (Error % = 0). Guarde como smoke.jmx.



Test Plan

Thread Group

HTTP Request Defaults

GET Movies

View Results Tree

Summary Report

Aggregate Report

Aggregate Report

Name:Aggregate Report

Comments:

Write results to file / Read from file

Filename

Browse...

Log/Display Only:

Errors

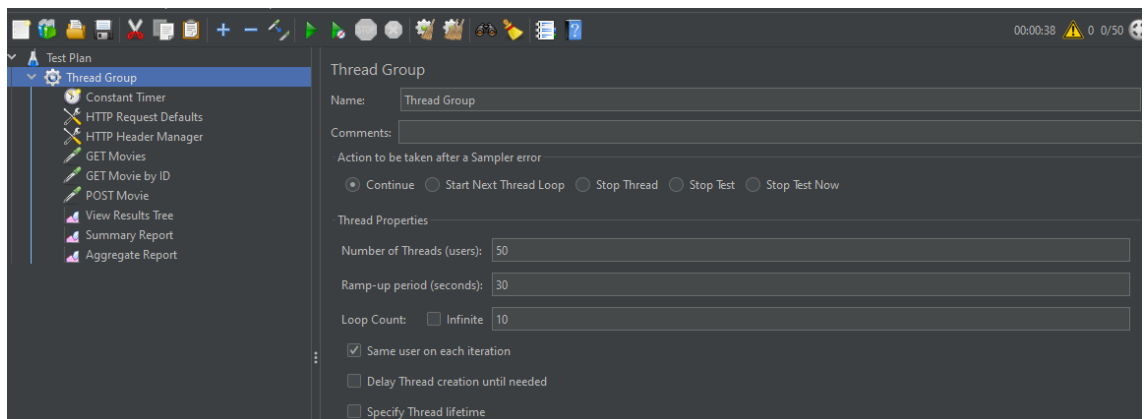
Successes

Configure

Label	# Samples	Average	Median	90% Line	95% Line	99% Line	Min	Maximum	Error %	Through...	Receive...	Sent KB/...
GET Mo...	5	12	3	4	50	50	3	50	0.00%	79.4/sec	451.47	9.53
TOTAL	5	12	3	4	50	50	3	50	0.00%	79.4/sec	451.47	9.53

Parte 3 — Prueba de carga

1. Duplique el plan. Configure el Thread Group a 50 usuarios, ramp-up 30 s, loops 10.
2. Agregue al menos 3 peticiones distintas: GET /movies, GET /movies/{id} (con CSV Data Set Config de IDs) y POST /movies.
3. Agregue un Constant Timer o Gaussian Random Timer (p. ej. 300 ms) para simular usuarios reales.
4. Ejecute la prueba y registre throughput, media, percentil 90/95 y Error % de cada petición.
5. Guarde como carga.jmx y exporte los resultados a carga.jtl.



Thread Group

Name: Thread Group

Comments:

Action to be taken after a Sampler error

☒ Continue ☐ Start Next Thread Loop ☐ Stop Thread ☐ Stop Test ☐ Stop Test Now

Thread Properties

Number of Threads (users): 50

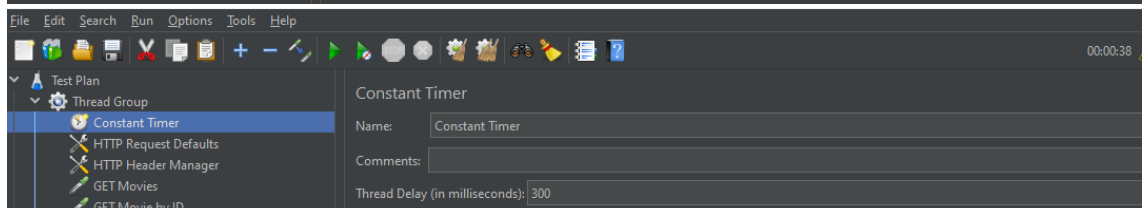
Ramp-up period (seconds): 30

Loop Count: ☐ Infinite 10

☒ Same user on each iteration

☐ Delay Thread creation until needed

☐ Specify Thread lifetime

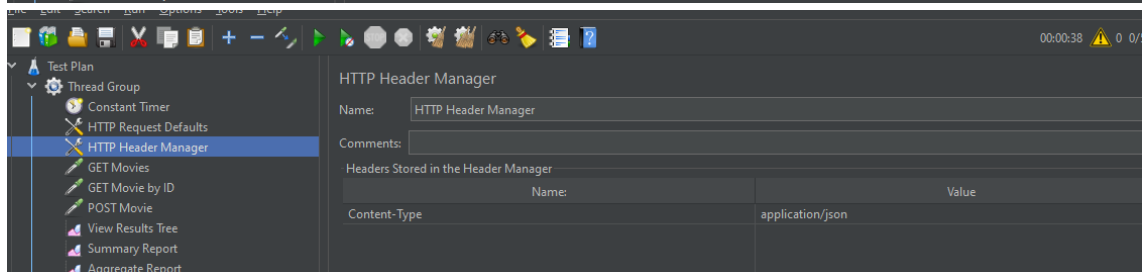


Constant Timer

Name: Constant Timer

Comments:

Thread Delay (in milliseconds): 300



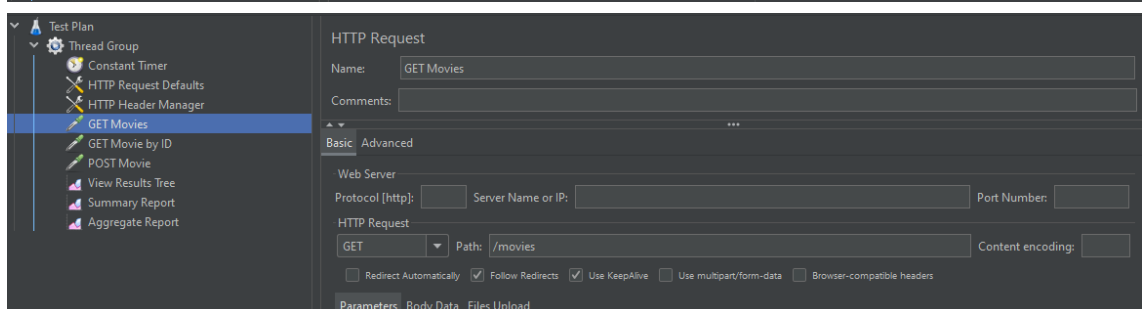
HTTP Header Manager

Name: HTTP Header Manager

Comments:

Headers Stored in the Header Manager

Name	Value
Content-Type	application/json



HTTP Request

Name: GET Movies

Comments:

Basic Advanced

Web Server

Protocol [http]: Server Name or IP: Port Number:

HTTP Request

GET Path: /movies Content encoding:

☐ Redirect Automatically ☒ Follow Redirects ☒ Use KeepAlive ☐ Use multipart/form-data ☐ Browser-compatible headers

Parameters Body Data Files Upload

Test Plan

- Thread Group
 - Constant Timer
 - HTTP Request Defaults
 - HTTP Header Manager
 - GET Movies
 - GET Movie by ID**
 - POST Movie
 - View Results Tree
 - Summary Report
 - Aggregate Report

HTTP Request

Name: GET Movie by ID

Comments:

Basic Advanced

Web Server

Protocol [http]: Server Name or IP: Port Number:

HTTP Request

GET Path: /movies/c7f00a84-7a22-458a-b3f7-9b3a688a2217 Content encoding:

☐ Redirect Automatically ☒ Follow Redirects ☒ Use KeepAlive ☐ Use multipart/form-data ☐ Browser-compatible headers

Parameters Body Data Files Upload

Send Parameters With the Request:

Name:	Value	URL Encode?	Content-Type	Include Equals?
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Test Plan

- Thread Group
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 - GET Movies
 - GET Movie by ID
 - POST Movie**
 - View Results Tree
 - Summary Report
 - Aggregate Report

HTTP Request

Name: POST Movie

Comments:

Basic Advanced

Web Server

Protocol [http]: Server Name or IP: Port Number:

HTTP Request

POST Path: /movies Content encoding:

☐ Redirect Automatically ☒ Follow Redirects ☒ Use KeepAlive ☐ Use multipart/form-data ☐ Browser-compatible headers

Parameters Body Data Files Upload

Body Data

```
1 {
2   "title": "JMeter Test Movie",
3   "director": "Nauren",
4   "genre": "sci-fi",
5   "year": 2026,
6   "rating": 8.5,
7   "synopsis": "Pellicula creata per JMeter"
8 }
```

Test Plan

- Thread Group
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 - HTTP Request Defaults
 - HTTP Header Manager
 - GET Movies
 - GET Movie by ID
 - POST Movie
 - View Results Tree**
 - Summary Report
 - Aggregate Report

View Results Tree

Name: View Results Tree

Comments:

Write results to file / Read from file

Filename: Browse... Log/Display Only: ☐ Errors ☐ Successes Configure

Search: ☐ Case sensitive ☐ Regular exp. Search Reset

Text

- GET Movie by ID
- POST Movie
- GET Movie by ID
- GET Movie by ID
- GET Movies
- POST Movie
- GET Movies
- POST Movie
- GET Movie by ID
- GET Movies
- POST Movie
- GET Movie by ID
- GET Movies
- POST Movie
- GET Movie by ID
- POST Movie
- GET Movies
- POST Movie

Sampler result Request Response data

Response Body Response headers

Find ☐ Case sensitive ☐ Regular exp.

Test Plan

- Thread Group
 - Constant Timer
 - HTTP Request Defaults
 - HTTP Header Manager
 - GET Movies
 - GET Movie by ID
 - POST Movie
 - View Results Tree
 - Summary Report**
 - Aggregate Report

Summary Report

Name: Summary Report

Comments:

Write results to file / Read from file

Filename: Browse... Log/Display Only: ☐ Errors ☐ Successes Configure

Label	# Samples	Average	Min	Max	Std. Dev.	Error %	Throughput	Received K...	Sent KB/sec	Avg. Bytes
GET Movies	500	8	2	66	4.92	0.00%	13.1/sec	864.02	1.99	67407.5
GET Movie...	500	3	1	59	3.43	0.00%	13.1/sec	7.06	2.46	550.0
POST Movie	500	10	3	77	6.78	0.00%	13.1/sec	6.34	4.36	494.0
TOTAL	1500	7	1	77	5.96	0.00%	38.7/sec	863.03	8.66	22817.2

Test Plan

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 - Aggregate Report

Summary Report

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Comments:

Write results to file / Read from file

Filename: Log/Display Only: ☐ Errors ☐ Successes

Label	# Samples	Average	Min	Max	Std. Dev.	Error %	Throughput	Received K...	Sent KB/sec	Avg. Bytes
GET Movies	2000	2360	11	4793	1135.69	0.00%	17.2/sec	10478.12	2.61	622026.8
GET Movie...	2000	2339	1	4579	1124.25	0.00%	17.3/sec	9.28	3.24	550.0
POST Movie	2000	2589	4	5165	1170.07	0.00%	17.3/sec	8.33	5.74	494.0
TOTAL	6000	2429	1	5165	1149.06	0.00%	51.5/sec	10439.88	11.51	207690.3

Test Plan

- Thread Group
 - Constant Timer
 - HTTP Request Defaults
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 - GET Movie by ID
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 - View Results Tree
 - Summary Report
 - Aggregate Report

Aggregate Report

Name: Aggregate Report

Comments:

Write results to file / Read from file

Filename: Log/Display Only: ☐ Errors ☐ Successes

Label	# Samples	Average	Median	90% Line	95% Line	99% Line	Min	Maximum	Error %	Through...	Receive...	Sent KB/...
GET Mo...	2000	2360	2551	3666	3843	4198	11	4793	0.00%	17.2/sec	10478.12	2.61
GET Mo...	2000	2339	2512	3620	3787	4192	1	4579	0.00%	17.3/sec	9.28	3.24
POST M...	2000	2589	2724	4016	4242	4873	4	5165	0.00%	17.3/sec	8.33	5.74
TOTAL	6000	2429	2599	3757	4008	4517	1	5165	0.00%	51.5/sec	10439.88	11.51

Etapas 3

Test Plan

- Thread Group
 - Constant Timer
 - HTTP Request Defaults
 - HTTP Header Manager
 - GET Movies
 - GET Movie by ID
 - POST Movie
 - View Results Tree
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Thread Group

Name: Thread Group

Comments:

Action to be taken after a Sampler error

☒ Continue ☐ Start Next Thread Loop ☐ Stop Thread ☐ Stop Test ☐ Stop Test Now

Thread Properties

Number of Threads (users):

Ramp-up period (seconds):

Loop Count: ☐ Infinite

Test Plan

- Thread Group
 - Constant Timer
 - HTTP Request Defaults
 - HTTP Header Manager
 - GET Movies
 - GET Movie by ID
 - POST Movie
 - View Results Tree
 - Summary Report
 - Aggregate Report

Summary Report

Name: Summary Report

Comments:

Write results to file / Read from file

Filename: Log/Display Only: ☐ Errors ☐ Successes

Label	# Samples	Average	Min	Max	Std. Dev.	Error %	Throughput	Received K...	Sent KB/sec	Avg. Bytes
GET Movies	4000	15043	66	27447	5501.92	0.00%	8.4/sec	11033.23	1.27	1351857
GET Movie...	4000	14895	3	27323	5460.56	0.00%	8.4/sec	4.49	1.57	550
POST Movie	4000	13955	5	28441	5724.81	0.00%	8.4/sec	4.03	2.78	494
TOTAL	12000	14631	3	28441	5584.49	0.00%	25.0/sec	11027.49	5.60	450967

Test Plan

- Thread Group
 - Constant Timer
 - HTTP Request Defaults
 - HTTP Header Manager
 - GET Movies
 - GET Movie by ID
 - POST Movie
 - View Results Tree
 - Summary Report
 - Aggregate Report

Aggregate Report

Name: Aggregate Report

Comments:

Write results to file / Read from file

Filename: Log/Display Only: ☐ Errors ☐ Successes

Label	# Samples	Average	Median	90% Line	95% Line	99% Line	Min	Maximum	Error %	Through...	Receive...	Sent KB/...
GET Mo...	4000	15043	14995	22234	24029	25865	66	27447	0.00%	8.4/sec	11033.23	1.27
GET Mo...	4000	14895	14701	22116	23903	25990	3	27323	0.00%	8.4/sec	4.49	1.57
POST M...	4000	13955	13435	21974	24469	27230	5	28441	0.00%	8.4/sec	4.03	2.78
TOTAL	12000	14631	14422	22100	24123	26215	3	28441	0.00%	25.0/sec	11027.49	5.60

Parte 5 — Prueba de picos y aserciones

1. Configure un escenario de pico: arranque brusco (ramp-up corto, p. ej. 200 usuarios en 5 s) y observe la recuperación.
2. Agregue una Response Assertion que valide código de respuesta 200 (o 201 en POST).
3. Agregue una Duration Assertion con un umbral acordado (p. ej. 800 ms) y observe cuántas peticiones lo incumplen bajo carga.
4. Guarde como picos.jmx.

En la prueba de picos con 200 usuarios y ramp-up de 5 segundos se observaron errores tipo Socket closed, indicando cierre de conexiones por saturación temporal del servidor/API bajo un incremento brusco de carga.